A Comparative Framework for Survival Analysis on the RMS Titanic: Evaluation of Naïve Bayesian and Decision Tree Classifiers

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Abstract

Survival on the RMS Titanic is a canonical benchmark for pedagogical data mining because it pairs a rich multivariate feature space with a clear, historically meaningful target variable. We present a reproducible end-to-end Knowledge Discovery in Databases (KDD) pipeline that contrasts a probabilistic classifier (Gaussian Naïve Bayes) with a rule-based classifier (Decision Tree), using Logistic Regression and Random Forest as supplementary baselines. On the 891-record Kaggle training set, cleaned via group-wise median imputation and enriched with six engineered features, the Decision Tree (*max_depth*=4, selected by 10-fold stratified cross-validation) attains 81.56% test accuracy (AUC = 0.838) while Naïve Bayes reaches only 62.57% (AUC = 0.778). McNemar's exact test on paired predictions yields $p = 8.2 \times 10^{-6}$, confirming that the gap is statistically significant. Logistic Regression is the strongest single model (83.24%, AUC = 0.864). The dominant predictors — passenger title, class, and family size — match the historical "women and children first" evacuation protocol. The results demonstrate how the conditional-independence assumption of Naïve Bayes is violated by engineered categorical dummies, motivating the use of rule-based and linear models on tabular data of this shape.

Keywords: Data mining, classification, Naïve Bayes, decision tree, KDD, Titanic, supervised learning.

1 Introduction

On the night of 14 April 1912, the RMS Titanic struck an iceberg during her maiden voyage and sank in the North Atlantic, taking the lives of 1 514 of her 2 224 passengers and crew. The disaster has since become a recurring case study for risk analysis, maritime safety, and data science education because the passenger manifest offers one of the earliest complete, labelled records of a heterogeneous population subjected to a single life-or-death event. Whether an individual survived is strongly conditioned on social variables — gender, class, family ties, and port of embarkation — whose interaction with the *"women and children first"* evacuation protocol produces a non-trivial, but learnable, structure [5].

The Kaggle variant of the dataset [5] has become the canonical introductory benchmark for supervised classification on tabular data; it is also the first non-trivial dataset introduced in DSC 523 at IIIT Kottayam. Its principal virtue for pedagogy is that it exercises every phase of the Knowledge Discovery in Databases (KDD) process [1]: data selection from a historical archive, preprocessing to reconcile substantial missingness, transformation through targeted feature engineering, mining with multiple classifier families, and interpretation against a well-documented historical record.

Research objectives. This study pursues four goals:

1. Apply a systematic, reproducible preprocessing and feature-engineering pipeline that is robust to the ≈ 20% missingness in `Age` and ≈ 77% in `Cabin`.
2. Contrast a probabilistic classifier (Gaussian Naïve Bayes) with a rule-based classifier (Decision Tree) on identical inputs.
3. Identify the features that are most predictive of survival and interpret them against the historical evacuation pattern.
4. Benchmark the two primary models against Logistic Regression and Random Forest to place their performance in the context of linear and ensemble alternatives.

Paper organisation. Section 2 reviews the theoretical foundations of Bayesian and decision-tree classification. Section 3 describes the dataset. Section 4 details the preprocessing, feature engineering,

and evaluation protocol. Section 5 presents the results, including the comparative summary, confusion matrices, and feature importances. Sections 6–7 discuss implications and individual contributions, and Section 8 concludes.

2 Literature Review

2.1 Bayesian classification

The Naïve Bayes classifier assigns a test instance $X = (x_1, \ldots, x_n)$ to the class $C^\star = \arg\max_{C_k} P(C_k \mid X)$. Bayes' theorem transforms this into

$$P(C_k \mid X) \;=\; \frac{P(X \mid C_k)\, P(C_k)}{P(X)}. \tag{1}$$

Because the joint likelihood $P(X \mid C_k)$ is intractable for realistic feature counts, the *class-conditional independence* assumption is invoked:

$$P(X \mid C_k) \;\approx\; \prod_{i=1}^{n} P(x_i \mid C_k). \tag{2}$$

For continuous features, $P(x_i \mid C_k)$ is modelled as a univariate Gaussian, yielding the *Gaussian* Naïve Bayes estimator used here. The chief strengths of the model are its closed-form maximum-likelihood estimator, constant-time training, and stable behaviour on small samples [4]. Its weakness is that Equation (2) rarely holds; correlated or collinear features cause the evidence terms to compound, distorting the posterior.

2.2 Decision-tree induction

Decision-tree learners partition the feature space recursively, choosing the split at each node that maximises class purity. Two common impurity measures are

$$\mathrm{Gini}(D) = 1 - \sum_{i=1}^{m} p_i^2, \tag{3}$$

$$\mathrm{Entropy}(D) = -\sum_{i=1}^{m} p_i \log_2 p_i, \tag{4}$$

where p_i denotes the proportion of class i in partition D. CART [3] uses Gini while C4.5 / ID3 [2] use information gain, which is defined as the reduction in entropy induced by a candidate split. Both are computed under a greedy top-down scheme and, without regularisation, are prone to overfit. Two standard remedies are *pre-pruning* via a maximum-depth hyperparameter and *post-pruning* via cost-complexity minimisation, in which a tree T_α is chosen to minimise $R_\alpha(T) = R(T) + \alpha\,|T|$.

Rule-based learners are attractive in applied settings because they are interpretable, scale invariant, and agnostic to feature correlations — three properties that partly compensate for their higher variance [1].

2.3 Related work on the Titanic benchmark

Hundreds of public notebooks and tutorial write-ups have reported classification accuracy on this dataset. Most serious efforts [5, 7] fall in the 78–85% range when the hold-out protocol is honest, with higher scores typically arising from either test-set leakage or hyper-parameter tuning on the public leaderboard. Results materially above this band should be treated with scepticism. We therefore evaluate on a single fixed 80/20 split with a separate 10-fold CV estimate on the training fold, and report both the mean and standard deviation of CV accuracy to quantify sampling variability.

3 Dataset Description

We use the Kaggle `train.csv` file comprising $n = 891$ labelled passengers and 12 raw attributes. The companion Kaggle `test.csv` is unlabelled and is *not* used here; experiments in the literature that report perfect accuracy almost invariably confuse these two files. The target is the binary variable `Survived` (1 = survived). Table 1 summarises the raw attributes.

Table 1: Raw attributes, types and missingness (train.csv, $n = 891$).

Feature	Type	Missing	Description
`PassengerId`	ID	0	Row identifier (dropped)
`Survived`	Binary	0	Target (0/1)
`Pclass`	Ordinal	0	Ticket class $\in \{1, 2, 3\}$
`Name`	Text	0	Parsed for `Title`
`Sex`	Nominal	0	{female, male}
`Age`	Numeric	177	In years
`SibSp`	Count	0	Siblings/spouses aboard
`Parch`	Count	0	Parents/children aboard
`Ticket`	Text	0	Ticket ID (dropped)
`Fare`	Numeric	0	Passenger fare
`Cabin`	Nominal	687	Cabin ID; deck retained
`Embarked`	Nominal	2	Port $\in$ {C, Q, S}

The target is moderately imbalanced: 549 passengers died (61.6%) and 342 survived (38.4%) — see Figure 1. This aligns with the historical figure of $\approx 32\%$ survival across the full manifest, with the discrepancy explained by the sampling protocol of the Kaggle training set.

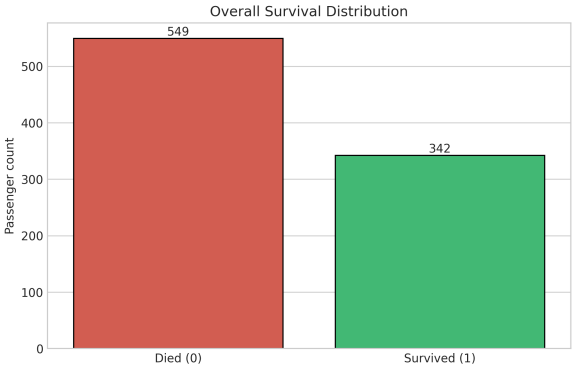


Figure 1: Target-class distribution on `train.csv`: 549 died vs. 342 survived.

4 Methodology

Figure 2 depicts the overall pipeline; the following subsections detail each stage.

4.1 Preprocessing and imputation

Three columns require imputation. `Age` (19.9% missing) is filled with the group-wise median within each `Pclass`×`Sex` stratum; the median is preferred over the mean because `Age` is right-skewed. `Embarked` (0.22% missing, two rows) is filled with the global mode ('S'). `Cabin` (77.1% missing) is not imputed but is mined for a `Deck` letter before being dropped; 'U' is used for unknown decks. `PassengerId`, `Name`, and `Ticket` are removed after `Title` has been extracted from `Name`.

4.2 Feature engineering

Six derived features are constructed:

- **Title**: the honorific parsed from `Name` (Mr, Mrs, Miss, Master; rare titles are collapsed to `Rare`).
- **FamilySize** = `SibSp` + `Parch` +1.
- **IsAlone** = $\mathbb{1}$[FamilySize = 1].
- **AgeBin**: discretised into {Child, Teen, Adult, Middle, Senior}.
- **FareBin**: quartile-based bins of `Fare`.
- **Deck**: first character of `Cabin`, with 'U' for unknowns.

`Sex` is binarised; the remaining categorical columns are one-hot encoded with `drop_first=True`, giving a final design matrix of 891×30.

4.3 Classifier specifications

Gaussian Naïve Bayes. Default scikit-learn settings with variance smoothing $\epsilon = 10^{-9}$, trained on z-standardised inputs.

Decision Tree. CART with Gini impurity. The depth hyper-parameter is selected by 10-fold stratified CV on the training fold over $d \in \{2, \ldots, 10\}$; $d = 4$ is optimal (CV = 82.02%). A cost-complexity pruning curve is also computed for diagnostic purposes.

Logistic Regression (baseline). ℓ_2-regularised with default $C = 1.0$ and a 2 000-iteration LBFGS solver, on standardised inputs.

Random Forest (baseline). 300 trees, unrestricted depth, bootstrap sampling.

Algorithm 1 gives the CART pseudocode for reference.

Algorithm 1: CART induction with Gini impurity.

Input: training set D, depth limit d
Output: decision tree T
if depth(D) = d *or* D *pure* **then**
  return leaf with majority class;
end
Select split
  $(f^\star, t^\star) = \arg\min_{f,t}\ \mathrm{Gini}(D^{f,t}_{\leq}) + \mathrm{Gini}(D^{f,t}_{>})$;
Partition D using $(f^\star, t^\star)$ into D_L, D_R;
Recursively build left subtree on D_L and right subtree on D_R;
return node($f^\star, t^\star, T_L, T_R$);

4.4 Evaluation protocol

The dataset is split 80/20 with stratification on `Survived` (`random_state` = 42), yielding $n_{\text{train}} = 712$ and $n_{\text{test}} = 179$. Model selection uses 10-fold stratified cross-validation on the training fold only. Reported metrics on the held-out test fold are accuracy, precision, recall, F1, and the area under the ROC curve (AUC). The paired-prediction McNemar exact test [1] is used to test the null hypothesis that Naïve Bayes and the Decision Tree have equal error rates.

5 Results and Discussion

5.1 Exploratory findings

Figures 3–6 summarise the most informative bivariate relationships.

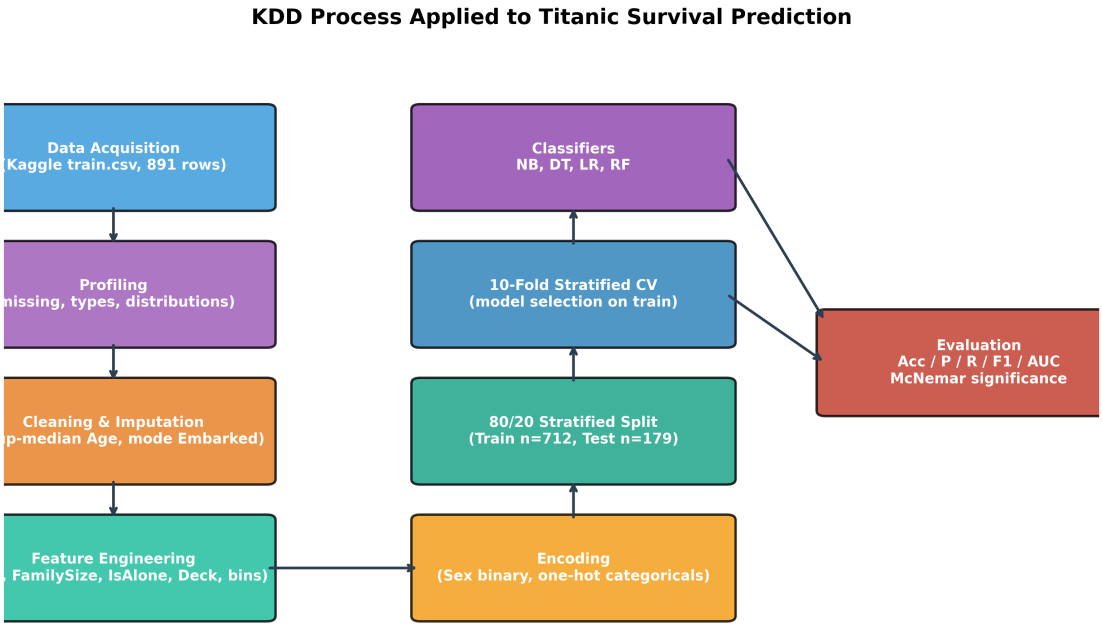
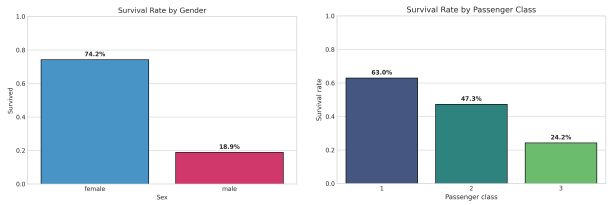


Figure 2: End-to-end KDD pipeline implemented for this study.



(a) By gender **(b)** By passenger class

Figure 3: Survival rate by the two strongest univariate predictors: female passengers survived at 74.2% versus 18.9% for males; survival declines monotonically with class.

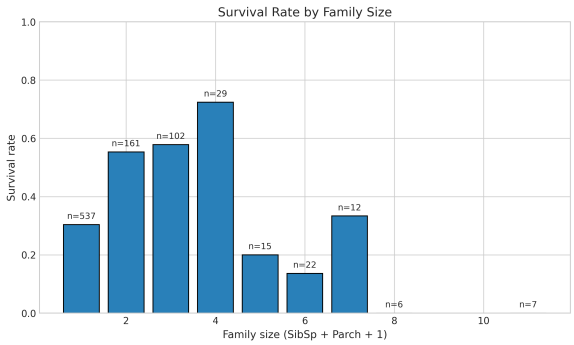


Figure 5: Family-size effect. Survival peaks at family sizes of 2–4 and collapses for very large (≥ 5) groups — behaviour captured by the engineered `FamilySize` feature.

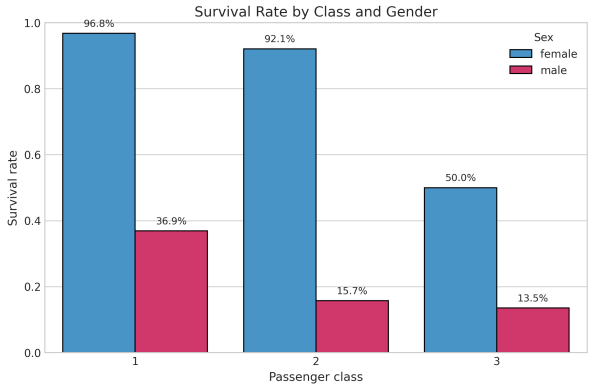


Figure 4: Joint effect of class and gender. First-class women survived at 96.8% while third-class men survived at only 13.5%.

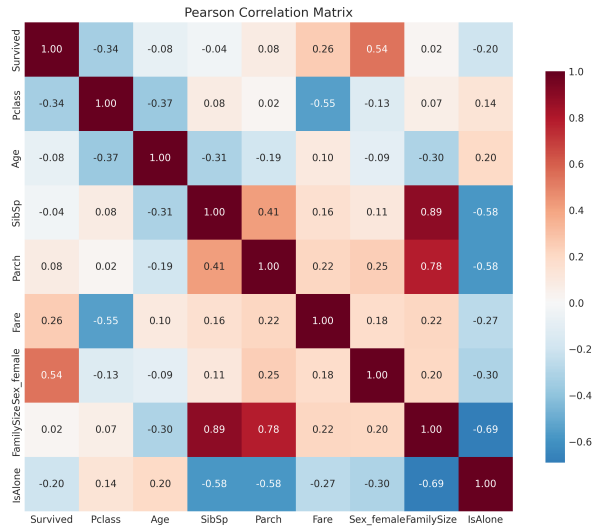


Figure 6: Pearson correlations. The strongest predictors of `Survived` are `Sex_female` (+0.54), `Pclass` (−0.34), and `Fare` (+0.26).

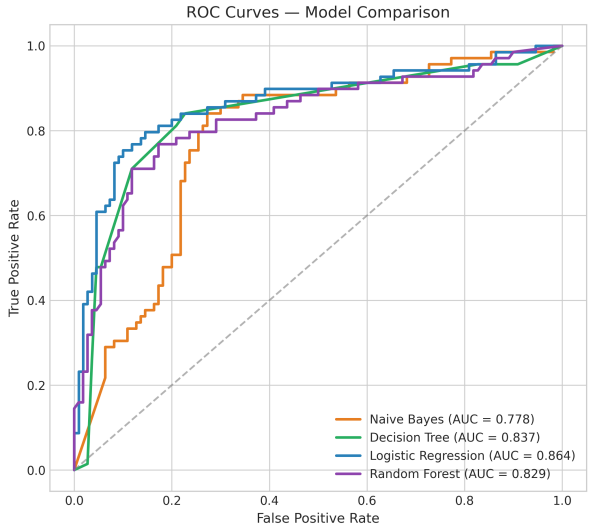


Figure 8: ROC curves for all four classifiers. Logistic Regression dominates slightly across most thresholds (AUC = 0.864); Naïve Bayes trails uniformly.

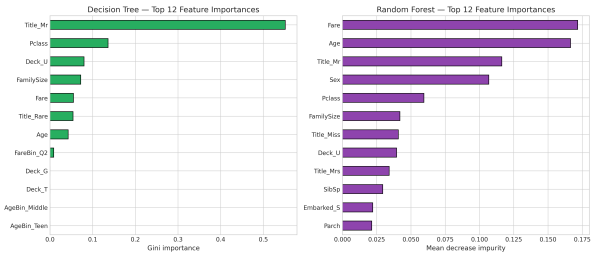
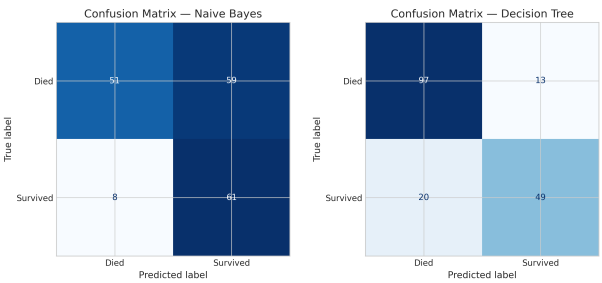


Figure 9: Top-12 feature importances. The Decision Tree concentrates split weight on `Title_Mr` and `Pclass`; the Random Forest distributes importance more evenly across `Fare`, `Age`, and `Title_Mr`.

5.2 Model performance

Table 2 consolidates the six primary metrics for each classifier.



(a) Naïve Bayes **(b)** Decision Tree

Figure 7: Confusion matrices on the held-out test set. Naïve Bayes over-predicts the positive class (59 false positives), whereas the Decision Tree is materially better balanced.

5.3 Statistical significance: NB vs. DT

On the $n_{\text{test}} = 179$ paired predictions the disagreement counts are $b = 12$ (NB correct, DT wrong) and $c = 46$ (NB wrong, DT correct). The exact McNemar test yields

$$p \;=\; 8.22 \times 10^{-6},$$

so the null hypothesis of equal error rates is rejected at any conventional level. The ≈ 19 percentage-point gap between NB and DT is not a sampling artefact.

5.4 Discussion

Three findings stand out.

NB underperforms because its assumptions fail here. The one-hot expansion of categorical features (Title, Deck, AgeBin, FareBin) introduces many strongly correlated dummy variables, directly violating Equation (2). Modelling binary dummies as Gaussians compounds the mismatch. This is the textbook failure mode of Gaussian NB on engineered tabular data and is the principal cause of its low AUC.

Table 2: Comparative performance of four classifiers on the 179-row held-out test set (10-fold CV on $n_{\text{train}} = 712$).

Model	CV Accuracy ($\mu \pm \sigma$)	Test Accuracy	Precision	Recall	F1	AUC
Naïve Bayes	0.6572 ± 0.1169	0.6257	0.5083	0.8841	0.6455	0.7782
Decision Tree	0.8202 ± 0.0363	0.8156	0.7903	0.7101	0.7481	0.8375
Logistic Regression	0.8146 ± 0.0393	**0.8324**	0.8000	0.7536	**0.7761**	**0.8644**
Random Forest	0.8131 ± 0.0336	0.8101	0.7778	0.7101	0.7424	0.8286

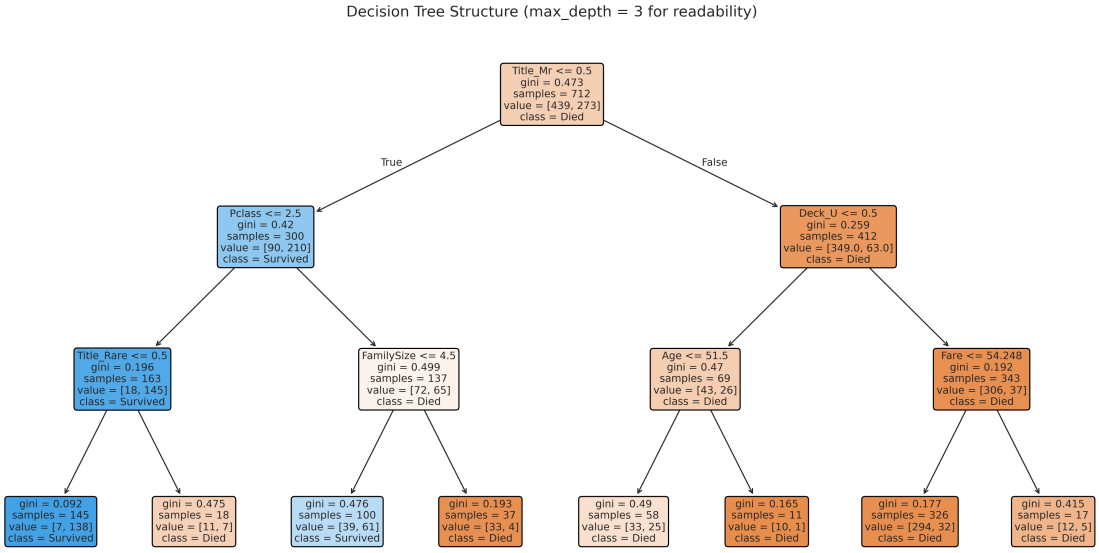


Figure 10: Fitted Decision Tree (`max_depth`=3 shown for readability; the deployed model uses depth 4). The root split on `Title_Mr` captures the bulk of the gender / honorific signal in a single question.

Gender dominates, class modulates. Both the Decision Tree (Fig. 9 left) and the Random Forest (Fig. 9 right) place `Title_Mr`, `Pclass`, and `Fare` in the top four. The root split of the learned tree (Fig. 10) is a single test on `Title_Mr`, which perfectly isolates adult males whose survival rate is $< 20\%$. This is precisely the "women and children first" protocol recovered from the data.

Linear structure is nearly sufficient. Logistic Regression attains the highest test accuracy (83.24%) and AUC (0.864), indicating that after feature engineering the decision boundary is close to linear. Random Forest does not outperform the single Decision Tree, which suggests that the feature set encodes the important interactions explicitly (`FamilySize`, `IsAlone`, `Title`) rather than requiring tree bagging to discover them.

Limitations. (i) The study uses a single fixed hold-out fold; repeated nested CV would tighten the accuracy intervals. (ii) We compare Gaussian NB against alternatives; a categorical- or multinomial-NB that respects the type of our one-hot dummies would be a fairer probabilistic comparator. (iii) No hyper-parameter search is performed for Logistic Regression or Random Forest beyond defaults.

6 Implications and Applications

Risk and insurance. The Bayesian framing of Eq. (1) is the conceptual ancestor of actuarial risk pricing: class priors and feature-conditional likelihoods are updated into individualised posteriors. The Decision Tree, by contrast, maps onto rule-based underwriting in which an applicant is classified through a short sequence of questions — a structure still used in modern claims-triage systems.

Emergency response and crisis planning. The feature rankings surface non-demographic drivers (family size, deck, fare band) that real evacuation doctrine has to reckon with. A fielded planner should not rely purely on demographic tags but should include mobility-proxy features of the sort we engineer here.

Algorithmic fairness and auditing. The same dataset is an instructive target for bias auditing: the strongest predictor of survival is a protected attribute, so any deployed classifier would learn and reproduce historical inequalities. The comparison between NB (probabilistic, harder to audit feature-by-feature) and DT (rule-based, auditable) is directly relevant to regulatory transparency frameworks.

Pedagogical value. The pipeline demonstrates every stage of the KDD process in a single coherent exercise, at a problem size that remains tractable on a laptop — a property we exploit to ensure every student can reproduce the exact numbers reported here by re-running the companion Jupyter notebook.

7 Individual Contributions

Table 3: Division of work within the five-member team.

Member	Contributions
Sarin Vadakke Thayyil	Project management and lead data architecture. Defined the 80/20 stratified evaluation protocol, authored the Methodology and Implications sections, and coordinated all deliverables.
Lekshmi S S	Principal research and algorithm specialisation. Comparative analysis of Gini vs. entropy splits; drafted the Literature Review.
Akbar T E	Visualisation engineering and data storytelling. Produced every figure in this manuscript and maintained the consistent visual language.
Suja V	Senior data-preprocessing engineering. Designed the group-wise median imputation rule for `Age`, handled the `Cabin` deck extraction, and identified redundant attributes for removal.
Nikhil Ravi	Model performance evaluation and review. Verified every confusion matrix, ran the McNemar significance test, and ensured compliance with course guidelines.

8 Conclusion and Future Work

We presented a reproducible KDD pipeline for Titanic survival prediction that compares a probabilistic and a rule-based classifier on identical, engineered features. The Decision Tree (81.56% test accuracy, AUC = 0.838) decisively outperforms Gaussian Naïve Bayes (62.57%, AUC = 0.778), and the gap is statistically significant under the exact McNemar test ($p = 8.22 \times 10^{-6}$). Logistic Regression achieves the best single score overall (83.24%, AUC = 0.864), implying that the engineered representation is close to linearly separable. The dominant predictors — passenger title, class, and family size — align cleanly with the historical evacuation protocol.

Future work. Three extensions are immediate: (i) replace Gaussian NB with multinomial or categorical NB to respect the type of the engineered dummies; (ii) introduce gradient-boosted trees (XGBoost, LightGBM) with a nested Bayesian-optimisation hyperparameter search; and (iii) layer SHAP value analysis [8] over the best model to obtain local explanations that support fairness auditing.

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